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Dear Editors,

I wish to submit the original research note entitled “When Every Parenthesis Matters” for consideration in *The American Mathematical Monthly*.

I confirm that this work is original, has not been published elsewhere, and is not currently under consideration for publication elsewhere.

The note asks which fully parenthesized expressions built from addition, multiplication, and division have the property that every inner pair of parentheses is mathematically necessary. It gives a local classification of necessary parentheses, derives a recursive structural description, and obtains the Fibonacci enumeration $r_2 = 3$ and $r_n = 2F_{n+1}$ for $n \geq 3$.

I believe the manuscript fits the aims and scope of the *Monthly* because it treats an elementary and immediately recognizable question through a short, self-contained argument connecting arithmetic notation, binary trees, and Fibonacci numbers. The result is intended to be accessible to a broad mathematical readership and suitable for discussion in undergraduate combinatorics or discrete mathematics.

Keywords: redundant parentheses; arithmetic expressions; binary trees; Fibonacci numbers; operator precedence.

I have no conflicts of interest to disclose. No specific funding was received for this work. OpenAI ChatGPT (GPT-5.6 Thinking) was used to assist with English-language editing, organization, formatting, and manuscript-preparation review. I independently verified all mathematical statements, proofs, computations, and references and take full responsibility for the manuscript.

Please address all correspondence concerning this manuscript to me at growup.kuriyama@gmail.com.

Thank you for your consideration.

Sincerely,

Takayuki Kuriyama